SEEKING SIXTH-GEN COMPANY

The future of aerial combat lies in **sixth generation fighter aircraft** acting as command nodes in an interconnected battlespace. India could get ahead of the game by partnering such an avant-garde jet programme

By **PRADIP R. SAGAR**

Aerial warfare is no longer what it used to be. Layered air defence networks, fighter jets with electronic warfare (EW) systems, seamless networking between ground-based, airborne and fighter-based radars, along with the inexpensive but disruptive unmanned aerial vehicles (UAVs) or drones have changed the nature of the game. Wars of the future will hinge on greater unification of all these through stealth and data fusion, as well as the ability to outmanoeuvre similar rival networks. Enter sixth-generation fighters, which are being conceived precisely for this reality—less as standalone jets and more as command nodes in an interconnected combat ecosystem. These jets are expected to include artificial intelligence (AI)-driven combat systems, manned-unmanned teaming (pairing with drones), long-range precision strike capabilities and advanced propulsion technologies. Two sixth-gen fighter programmes have caught the world's attention: America's F-47 under its Next Generation Air Dominance (NGAD) programme and China's J-36 and J-50 projects.

Now, India has an opportunity to join a sixth-gen fighter jet programme, with the Union Ministry of Defence informing Parliament in March 2026 of such an intent. The choice is between two other sixth-gen projects—the Future Combat Air System (FCAS) originally led by France, Germany and Spain (though France has recently pulled out of the project), and the Global Combat Air Programme (GCAP) of the United Kingdom, Italy and Japan. On April 30, defence secretary Rajesh Kumar Singh had said, "We have approached both ongoing partnerships between allied democracies and expressed interest in joining a sixth-generation fighter programme. The response is awaited."

Air Marshal Raghunath Nambiar, former deputy chief of the Indian Air Force (IAF), believes that India's participation in a sixth-gen fighter programme would be transformative. For India, it offers access to cutting-edge technologies in AI, engine technology and networked warfare, while enhancing deterrence against adversaries. For Europe, it provides shared development costs and an



GCAP#

- A collaborative effort between the United Kingdom, Italy and Japan, the Global Combat Air Programme (GCAP) was formed in late 2022 by merging the UK-led Tempest project with Japan's F-X programme
- **MAIN FEATURES:** Emphasis on digital design, use of virtual reality and 3D modelling to accelerate flight testing phase. Integration of Japan's advanced radar technology with the UK's propulsion expertise
- A 'system-of-systems' approach to develop a highly-networked, adaptable combat ecosystem with AI-driven combat systems, pairing with drones and long-range precision strike capabilities
- **KEY PARTNERS:** BAE Systems (UK), Leonardo (Italy) and Mitsubishi Heavy Industries (Japan)
- **ENGINE PARTNERS:** Rolls-Royce (UK), Avio Aero (Italy) and IHI Corporation (Japan)
- **TARGET INDUCTION:** 2035